ABDURAIMJONOV
ABDURAKHMON

AI Engineer and Software Developer

With 5 years in AI and Computer Vision, I specialize in object classification, detection, pose estimation, and image processing. I also have 3 years of experience in automation testing for Android and iOS, as well as developing GUIs for kiosks and mobile platforms. Additionally, I am skilled in socket and serial connections with hardware devices and have a knowladge working of databases. *Please visit the **Portfolio link** to see the results of my researches.*

Portfolio link: https://aavuzb.github.io/Portfolio/

Experience

SMHACCP Osan, Korea — *February 27, 2024– Present*

Senior Researcher | Machine Vision and AI R&D Team

UI apps: Label Printer, Data Collect&Insert data into Data Bases

Socket&Serial connections: PLC, Scalers, Printers, RS232 and so on. Get info from hardware devices and insert into Data Bases

RISO Convergence Seoul, Korea — *Januaey 2024– February 2024*

Senior Researcher | AI R&D Team

- **Motion Capture:** Connecting **3D** human models with **Pose estimation** models to make animations, movies

ONYCOM Inc. Seoul, Korea — *November 2020 – December 2023*

Senior Researcher | Solution Team

- **Automation Tool** to **test IOS** and **Android apps** automatically
- Implementing Object **Detection Model**s to detect **all objects** on **Mobile applications** (e.g. Icon, Image, Text ...)
- New **LabelImg** tool for **Rule Based testing**, **Auto Labeling**, **Auto Data Augmentation** and **Auto Training**
- **New Multi-Scale Template Matching** method to **control** and **test multi-devices** in the **same time**
- **Rest APIs** for Detection Models and Automation tools

GREW CREATIVE LAB Incheon, Korea — *June 2020 – October 2020*

Junior Researcher | R&D Team

- Implementing **Software Application** to combine **Unity 3D** and **Mujoco Physics**
- **Reinforcement Learning** researches: Robotics, Mujoco, Q-Learning, Actor-Critic, EPOpt and ML-Agents

ROBORUS Seoul, Korea — *July 2019 – June 2020*

Junior Researcher | R&D Team

- **Café cleanliness** (**Object Recognition** project)
- **Self-Ordering** applications for **Kiosk** and **Mobile** devices
- **Face** and **Emotion detection** models and **APIs**

CVPR LAB Kumoh National Institute of Technology — *2017.9 – 2019.8*

Researcher | Master's degree student

- **Image quality** improvement: removing fog, haze and noise
- Improving Classification result using **Fast Fourier Transform**
- **Car toys** recognition and detection

Contact

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Seoul, Republic of Korea

About Me

Birth year: **1994**

Nationality: **UZBEKISTAN**

Visa Type: **F-5**

Languages skills

Uzbek

English

Korean

Interested in

- Artificial Intelligence
- Graphical User Interface
- Computer Vision
- Mathematics
- Medical IT
- Robotics
- Automation Tasks

Hobbies

- Playing soccer
- Traveling
- Watch comedy & action movies

Knowledge

- Artificial Intelligence
  - Machine Learning
  - Deep Learning
  - Reinforcement Learning
- Computer Vision, Machine Vision
- Graphical User Interface
- Web development
- Blender, Unity 3D
- Automation Testing mobile android&ios apps
- Operation Systems
  - Linux
  - Mac
  - Windows

Programming Languages

Python

Tkinter, OpenCV, Numpy, PyTorch, Keras, Tensorflow, Matplotlib, Flask, Django, Json, XML, UI Automators, Scikit-learn, Requests, Pandas, WebDriveAgent, Blender

C++

OpenCV, Openframeworks, C++ Builder

JavaScript

HTML, CSS

C#

Unity

R

Strength

> Innovative, Critical observation
> Patience, Smart work
> Quick learner, Motivator
> Helping, Friendly

Education

Master's degree in **Computer Engineering**, Kumoh National Institute of Technology

GPA: **4.1** / 4.5
Common Courses: Artificial Intelligence, Comput Vision, Image Processing, Machine Learning and Deep Learning
Period: **2017-2019**
Thesis: Concatenation of Spatial and Frequency image for recognition of roughly detected characters

Bachelor's degree in **Computer Engineering**, Tashkent University of Information Technologies

GPA: **75.1** / 100
Common Courses: – Linear Algebra, Calculus, Probability and Statistics, Programming in C++/Java, Web Application Development in JavaScript, WordPress
Period: **2013-2017**
Graduation Project: Online market for selling applications

Publications

Patent:

- 딥러닝 기술과 새로운 다중 스케일의 템플릿 매칭 기술 융합을 통한 특정 객체탐지 기술 (Specific object detection technology through convergence of deep learning technology and new multi-scale template matching technology) **2022**

Papers:

- Extending Input Channel Using Global Feature Image for Convolutional Neural Networks
  https://ieeexplore.ieee.org/document/8960966 **2019**
- INNOVATIVE INFORMATION TECHNOLOGY IN EDUCATION
  https://forum-nauka.ru/domains_data/files/9/Karimov.pdf **2017**

Language Certificates

KIIP Level 5 (사회통합프로그램 5 단계 힙격) April 26, **2023**
IELTS 5.5 band of April, **2017**

Reference

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UDEMY Python Course instructor

- Making **Basic Python Programming** course. Course name is **Python for Beginners. Learn Twice (FREE Presentation EBOOK)**
  Link: https://www.udemy.com/course/twice-learning-python-programming-language/?src=sac&kw=learn+twice